Q1 2021 (01 Sep Shift 2)

The numbers of pairs (a, b) of real numbers, such that whenever α is a root of the equation $x^2 + ax + b = 0, \alpha^2 - 2$ is also a root of this equation, is :

(1) 6

(2) 2

(3) 4

(4) 8

Q2 2021 (31 Aug Shift 1)

cosec 18° is a root of the equation :

(1) $x^2 + 2x - 4 = 0$

(2) $4x^2 + 2x - 1 = 0$

(3) $x^2 - 2x + 4 = 0$

(4) $x^2 - 2x - 4 = 0$

Q3 2021 (31 Aug Shift 1)

The number of real roots of the equation $\mathrm{e}^{4\mathrm{x}} + 2\mathrm{e}^{3\mathrm{x}} - \mathrm{e}^{\mathrm{x}} - 6 = 0$ is:

(1) 2

(2) 4

(3) 1

(4) 0

Q4 2021 (27 Aug Shift 2)

The set of all values of $\mathrm{k} > -1$, for which the equation $\left(3x^2 + 4x + 3\right)^2 - (k+1)\left(3x^2 + 4x + 3\right)\left(3x^2 + 4x + 2\right) + k\left(3x^2 + 4x + 2\right)^2 = 0$ has real roots is :

(1) $\left(1, \frac{5}{2}\right]$

(2) $[2, 3)$

(3) $\left[-\frac{1}{2}, 1\right)$

(4) $\left(\frac{1}{2}, \frac{3}{2}\right] - \{1\}$

Q5 2021 (27 Aug Shift 1)

The number of distinct real roots of the equation $3x^4 + 4x^3 - 12x^2 + 4 = 0$ is .

Q6 2021 (26 Aug Shift 2)

Let $\lambda \neq 0$ be in $\mathbf{R}$. If α and β are the roots of the equation $x^2 - x + 2\lambda = 0$, and α and γ are the roots of equation $3\mathrm{x}^2 - 10\mathrm{x} + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to .

Q7 2021 (26 Aug Shift 1)

The sum of all integral values of $\mathrm{k}(\mathrm{k} \neq 0)$ for which the equation $\frac{2}{x-1} - \frac{1}{x-2} = \frac{2}{k}$ in x has no real roots, is .

Answer Key

Q1 (1)	**Q2** (4)	**Q3** (3)	**Q4** (1)
Q5 (4)	**Q6** (18)	**Q7** (66)	

Q1 (1)

Consider the equation $x^2 + ax + b = 0$

If has two roots (not necessarily real $\alpha \& \beta$)

Either $\alpha = \beta$ or $\alpha \neq \beta$

$Case(1)$ If $\alpha = \beta$, then it is repeated root. Given that $\alpha^2 - 2$ is also a root

So, $\alpha = \alpha^2 - 2 \Rightarrow (\alpha + 1)(\alpha - 2) = 0$

$\Rightarrow \alpha = -1$ or $\alpha = 2$

When $\alpha = -1$ then $(a, b) = (2, 1)$

$\alpha = 2$ then $(a, b) = (-4, 4)$

$Case(2)$ If $\alpha \neq \beta$ Then

(I) $\alpha = \alpha^2 - 2$ and $\beta = \beta^2 - 2$

Here $(\alpha, \beta) = (2, -1)$ or $(-1, 2)$

Hence $(a, b) = (-(\alpha + \beta), \alpha\beta)$

$= (-1, -2)$

(II) $\alpha = \beta^2 - 2$ and $\beta = \alpha^2 - 2$

Then $\alpha - \beta = \beta^2 - \alpha^2 = (\beta - \alpha)(\beta + \alpha)$

Since $\alpha \neq \beta$ we get $\alpha + \beta = \beta^2 + \alpha^2 - 4$

$\alpha + \beta = (\alpha + \beta)^2 - 2\alpha\beta - 4$

Thus $-1 = 1 - 2\alpha\beta - 4$ which implies

$\alpha\beta = -1$ Therefore $(\mathrm{a}, \mathrm{b}) = (-(\alpha + \beta), \alpha\beta)$

$= (1, -1)$

(III) $\alpha = \alpha^2 - 2 = \beta^2 - 2$ and $\alpha \neq \beta$

$\Rightarrow \alpha = -\beta$

Thus $\alpha = 2, \beta = -2$

$\alpha = -1, \beta = 1$

Therefore $(a, b) = (0, -4) \& (0, -1)$

(IV) $\beta = \alpha^2 - 2 = \beta^2 - 2$ and $\alpha \neq \beta$ is same as (III)

Therefore we get 6 pairs of (a, b)

Which are $(2,1),(-4,4),(-1,-2),(1,-1)(0,-4)$

Option (1)

Q2 (4)

$\operatorname{cosec} 18^\circ = \frac{1}{\sin 18^\circ} = \frac{4}{\sqrt{5}-1} = \sqrt{5}+1$

Let $\operatorname{cosec} 18^\circ = x = \sqrt{5}+1$

$\Rightarrow x-1=\sqrt{5}$

Squaring both sides, we get

$x^2-2x+1=5$

$\Rightarrow x^2-2x-4=0$

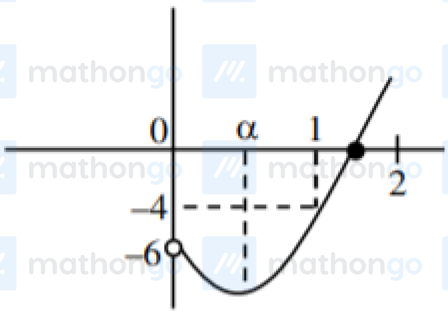
Q3 (3)

Let $\mathrm{e}^{\mathrm{x}} = \mathrm{t} > 0$

$f(\mathrm{t}) = \mathrm{t}^4 + 2\mathrm{t}^3 - \mathrm{t} - 6 = 0$

$f'(\mathrm{t}) = 4\mathrm{t}^3 + 6\mathrm{t}^2 - 1$

$f''(\mathrm{t}) = 12\mathrm{t}^2 + 12\mathrm{t} > 0$



$f(0)=-6, f(1)=-4, f(2)=24$

$\Rightarrow$ Number of real roots $=1$

Q4 (1)

$\left(3x^2+4x+3\right)^2-(k+1)\left(3x^2+4x+3\right)\left(3x^2+4x+2\right)$

$+k\left(3x^2+4x+2\right)^2=0$

Let $3x^2+4x+3=a$

and $3x^2+4x+2=b \Rightarrow b=a-1$

Given equation becomes

$\Rightarrow a^2-(k+1)ab+kb^2=0$

$\Rightarrow a(a-kb)-b(a-kb)=0$

$\Rightarrow (a-kb)(a-b)=0 \Rightarrow a=kb$ or $a=b($ reject $)$

$\because a=kb$

$\Rightarrow \quad 3x^2+4x+3=k\left(3x^2+4x+2\right)$

$\Rightarrow \quad 3(k-1)x^2+4(k-1)x+(2k-3)=0$

for real roots

$\mathrm{D} \geq 0$

$\Rightarrow 16(\mathrm{k}-1)^2-4(3(\mathrm{k}-1))(2\mathrm{k}-3) \geq 0$

$\Rightarrow 4(\mathrm{k}-1)\{4(\mathrm{k}-1)-3(2\mathrm{k}-3)\} \geq 0$

$\Rightarrow 4(\mathrm{k}-1)\{-2\mathrm{k}+5\} \geq 0$

$\Rightarrow -4(\mathrm{k}-1)\{2\mathrm{k}-5\} \geq 0$

$\Rightarrow (\mathrm{k}-1)(2\mathrm{k}-5) \leq 0$

$\therefore \mathrm{k} \in \left[1, \frac{5}{2}\right]$

$\because \mathrm{k} \neq 1$

$\therefore \mathrm{k} \in \left(1, \frac{5}{2}\right]$ Ans.

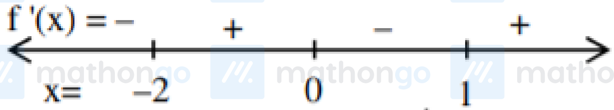
Q5 (4)

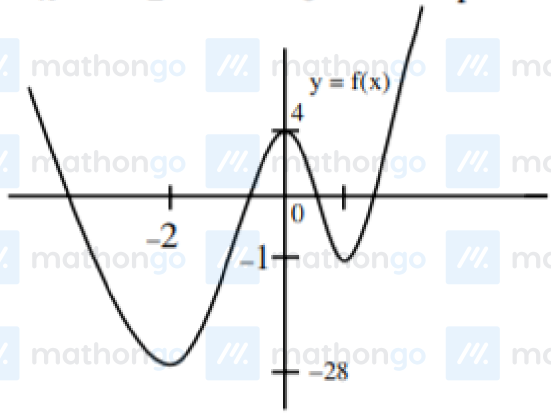
$3x^4 + 4x^3 - 12x^2 + 4 = 0$

So, Let $f(x) = 3x^4 + 4x^3 - 12x^2 + 4$

$\therefore \mathrm{f}'(\mathrm{x}) = 12\mathrm{x}\left(\mathrm{x}^2 + \mathrm{x} - 2\right)$

$= 12\mathrm{x}(\mathrm{x}+2)(\mathrm{x}-1)$





Q6 (18)

$3\alpha^2 - 10\alpha + 27\lambda = 0$ ____(1)

$\alpha^2 - \alpha + 2\lambda = 0$ ____(2)

(1) $-3(2)$ gives

$-7\alpha + 21\lambda = 0 \Rightarrow \alpha = 3\lambda$

Put $\alpha = 3\lambda$ in equation (1) we get

$9\lambda^2 - 3\lambda + 2\lambda - 0$

$9\lambda^2 = \lambda \Rightarrow \lambda = \frac{1}{9}$ as $\lambda \neq 0$

Now $\alpha = 3\lambda \Rightarrow \lambda = \frac{1}{3}$

$\alpha + \beta = 1 \Rightarrow \beta = 2/3$

$\alpha + \gamma = \frac{10}{3} \Rightarrow \gamma = 3$

$\frac{\beta\gamma}{\lambda} = \frac{\frac{2}{3}\times 3}{\frac{1}{9}} = 18$

Q7 (66)

$\frac{2}{x-1}-\frac{1}{x-2}=\frac{2}{k}$

$x \in R-\{1,2\}$

$\Rightarrow \mathrm{k}(2 \mathrm{x}-4-\mathrm{x}+1)=2\left(\mathrm{x}^{2}-3 \mathrm{x}+2\right)$

$\Rightarrow \mathrm{k}(\mathrm{x}-3)=2\left(\mathrm{x}^{2}-3 \mathrm{x}+2\right)$

for $x \neq 3, \quad k=2\left(x-3+\frac{2}{x-3}+3\right)$

$\mathrm{x}-3+\frac{2}{\mathrm{x}-3} \geq 2 \sqrt{2}, \forall \mathrm{x}>3$

$\& x-3+\frac{2}{x-3} \leq-2 \sqrt{2}, \forall x<-3$

$\Rightarrow 2\left(x-3+\frac{2}{x-3}+3\right) \in(-\infty, 6-4 \sqrt{2}] \cup[6+4 \sqrt{2}, \infty)$

for no real roots

$\mathrm{k} \in(6-4 \sqrt{2}, 6+4 \sqrt{2})-\{0\}$

Integral $\mathrm{k} \in\{1,2 \ldots \ldots 11\}$

Sum of $\mathrm{k}=66$